

Title: SMS Spam Detection

Abstract :

Spam SMS messages have become a major challenge in modern communication systems due to their increasing use in phishing attacks, financial fraud, malicious links, and unsolicited advertisements. Traditional machine learning techniques often rely on fixed features and may not effectively capture the different types of information present in SMS messages. Since different messages may contain different semantic meanings, textual patterns, and contextual information, an adaptive approach is required for effective spam detection.

The proposed project, "Dynamic Evidence Routing Transformer for Spam SMS Detection," aims to develop an intelligent and adaptive spam detection system using a Transformer-based architecture. The system utilizes an SMS dataset containing labeled SPAM and HAM (Non-Spam) messages. After preprocessing the SMS data and dividing it into training, validation, and testing sets, the proposed model processes the messages through three specialized components: Semantic Expert, Pattern Expert, and Entity/Context Expert. These experts capture the semantic meaning, local textual patterns, and contextual information present in SMS messages.

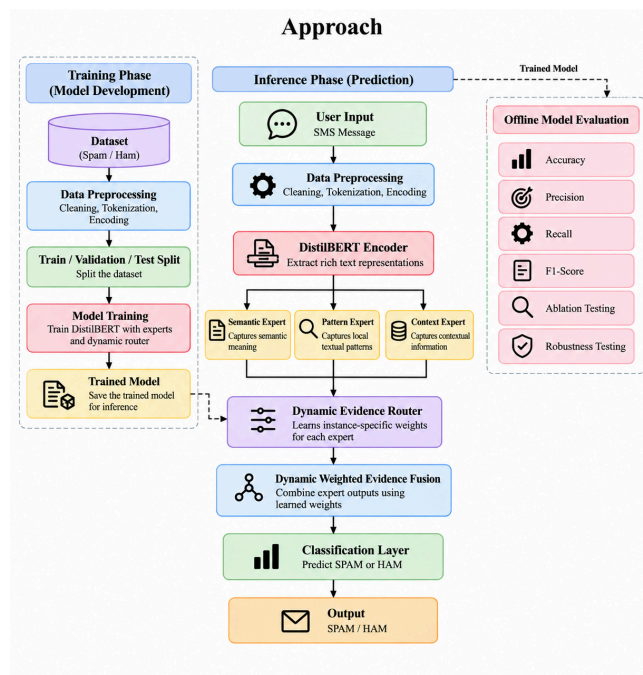
To improve the adaptability of the classification process, the proposed framework introduces a Dynamic Evidence Router that learns the contribution of each expert for an individual SMS message. The outputs from the three expert branches are dynamically weighted and combined before being passed to the classification layer. The system then classifies each message as either SPAM or HAM based on the combined evidence.

The performance of the proposed model is evaluated using standard metrics such as Accuracy, Precision, Recall, and F1-Score. In addition, ablation experiments are used to analyze the contribution of individual expert components and the dynamic routing mechanism, while robustness testing is performed to evaluate the model under challenging SMS variations. The proposed framework aims to provide an effective and adaptive spam detection solution by dynamically selecting and combining relevant textual evidence for reliable SMS classification.

Datasets

Dataset	URL Link
SMS Spam Collection Dataset	Kaggle
SMS Spam Collection (Hugging Face)	Hugging Face – Spam Detection Dataset
SMS Spam Collection (Hugging Face)	Hugging Face – UCI SMS Spam Dataset

Approach



Project Group Members

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